# Concavity, Curve Sketching, and Optimization

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Increasing/decreasing and the first derivative test**

For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where $f$ is increasing and decreasing, and classify each critical point using the First Derivative Test.

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. $f' > 0$ on $(-\infty, 1)$, $f' < 0$ on $(1, 3)$, $f' > 0$ on $(3, \infty)$. So $f$ is increasing on $(-\infty, 1) \cup (3, \infty)$, decreasing on $(1, 3)$. $x = 1$ is a local max (sign $+ \to -$); $x = 3$ is a local min (sign $- \to +$).

**2** **[Foundational]** • **Target: Concavity and the second derivative test**

For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

$f'(x) = 4x^3 - 12x^2$; $f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ on $(-\infty, 0)$, $f'' < 0$ on $(0, 2)$, $f'' > 0$ on $(2, \infty)$. Concave up on $(-\infty, 0) \cup (2, \infty)$, concave down on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$ (both sign changes).

**3** **[Foundational]** • **Target: Concavity and the second derivative test**

Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 3x$.

$f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$. $f''(x) = 6x$. $f''(1) = 6 > 0$: local min at $x = 1$. $f''(-1) = -6 < 0$: local max at $x = -1$.

**4** **[Foundational]** • **Target: Curve sketching**

For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.

$f'(x) = 3x^2 - 6x = 3x(x-2)$: critical points $x = 0, 2$. $f''(x) = 6x - 6$: zero at $x = 1$. Sign chart: $f' > 0$ on $(-\infty, 0)$, $f' < 0$ on $(0, 2)$, $f' > 0$ on $(2, \infty)$ — local max at $x = 0$ ($f(0) = 0$), local min at $x = 2$ ($f(2) = -4$). $f'' < 0$ on $(-\infty, 1)$ (concave down), $f'' > 0$ on $(1, \infty)$ (concave up) — inflection point at $x = 1$ ($f(1) = -2$).

**5** **[Foundational]** • **Target: Optimization problems**

A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.

Let $x$ = side parallel to river, $y$ = each perpendicular side. Constraint: $x + 2y = 200$, so $x = 200 - 2y$. Area $A = xy = (200 - 2y)y = 200y - 2y^2$. $A'(y) = 200 - 4y = 0 \Rightarrow y = 50$. $x = 200 - 100 = 100$. Maximum area occurs at $x = 100$ m, $y = 50$ m (verified max since $A'' = -4 < 0$).

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**6** [Foundational] • Target: Optimization problems

An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side $x$ from each corner and folding up the sides. Find $x$ that maximizes the volume.

$V(x) = x(12-2x)^2$ for $0 < x < 6$.
$V'(x) = (12-2x)^2 + x \cdot 2(12-2x)(-2) = (12-2x)[(12-2x) - 4x] = (12-2x)(12-6x)$. Setting $V' = 0$: $x = 6$ (rejected, gives $V = 0$) or $x = 2$. $V(2) = 2(8)^2 = 128$ in$^3$, the maximum.

**7** [Foundational] • Target: Optimization: minimizing cost or distance

Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

$D(x) = x^2 + (2x-3)^2 = x^2 + 4x^2 - 12x + 9 = 5x^2 - 12x + 9$. $D'(x) = 10x - 12 = 0 \Rightarrow x = \frac{6}{5}$.
$y = 2\left(\frac{6}{5}\right) - 3 = \frac{12}{5} - 3 = -\frac{3}{5}$. Closest point: $\left(\frac{6}{5}, -\frac{3}{5}\right)$ (confirmed min since $D'' = 10 > 0$).

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Optimization: minimizing cost or distance

A closed cylindrical can must have volume $16\pi$ cm$^3$. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes surface area.

From $V$: $h = \frac{16}{r^2}$. $S(r) = 2\pi r^2 + 2\pi r \cdot \frac{16}{r^2} = 2\pi r^2 + \frac{32\pi}{r}$. $S'(r) = 4\pi r - \frac{32\pi}{r^2} = 0 \Rightarrow 4\pi r = \frac{32\pi}{r^2} \Rightarrow r^3 = 8 \Rightarrow r = 2$ cm.

**9** [Intermediate] • Target: Points of inflection and graph behavior

Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

$f'(x) = 3x^2 \geq 0$ everywhere, so $f$ is never decreasing — $x = 0$ is not a local extremum. However $f''(x) = 6x$ changes sign at $x = 0$ (from concave down to concave up), so $x = 0$ IS an inflection point. This shows concavity change and extrema are independent features.

**10** [Intermediate] • Target: Visualizing extrema and concavity together

The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.

The graph rises to a peak at $x = 0$ ($f = 0$), then falls, changing its bend from concave-down to concave-up right at $x = 1$ ($f = -2$, the inflection point), before reaching a valley at $x = 2$ ($f = -4$) and rising again — visually confirming all three features found algebraically.

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**11** **[Intermediate]** • **Target: A capstone optimization problem**

A rectangular box with a square base and no top must have volume $500$ cm$^3$. Using $V = x^2h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length $x$ that minimizes the surface area.

$S(x) = x^2 + 4x \cdot \frac{500}{x^2} = x^2 + \frac{2000}{x}$. $S'(x) = 2x - \frac{2000}{x^2} = 0 \Rightarrow 2x^3 = 2000 \Rightarrow x^3 = 1000 \Rightarrow x = 10$ cm.

**12** **[Intermediate]** • **Target: Second derivative test failures**

For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.

$f'(x) = 4x^3 = 0$ gives $x = 0$. $f''(x) = 12x^2$; $f''(0) = 0$, so the Second Derivative Test gives no information. Using the First Derivative Test: $f'(x) < 0$ for $x < 0$ and $f'(x) > 0$ for $x > 0$, so $x = 0$ is a local (and in fact absolute) minimum.

**13** **[Intermediate]** • **Target: A capstone curve-sketching problem**

For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.

$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2) = 12x(x - 2)(x + 1)$: critical points $x = -1, 0, 2$.
$f''(x) = 36x^2 - 24x - 24$. $f''(-1) = 36 + 24 - 24 = 36 > 0$: local min. $f''(0) = -24 < 0$: local max.
$f''(2) = 144 - 48 - 24 = 72 > 0$: local min.

**14** **[Intermediate]** • **Target: Applied Optimization: Cost & Distance**

Find the nonzero inflection point of $f(x) = x^4 - 4x^3$.

Final Answer: $2$. $f''(x) = 12x^2 - 24x = 6x(2x - 4)$. Zero at $x = 0$ and $x = 2$; the nonzero inflection point is $x = 2$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Increasing/decreasing and the first derivative test** • **Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Find the nonzero inflection point of $f(x) = x^4 - 10x^3$.

Final Answer: $5$. $f''(x) = 12x^2 - 60x = 6x(2x - 10)$. Zero at $x = 0$ and $x = 5$; the nonzero inflection point is $x = 5$.

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**16** [Challenging] • Target: Concavity and the second derivative test • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find the nonzero inflection point of $f(x) = x^4 + 2x^3$.

Final Answer: $-1$. $f''(x) = 12x^2 + 12x = 6x(2x + 2)$. Zero at $x = 0$ and $x = -1$; the nonzero inflection point is $x = -1$.

**17** [Challenging] • Target: Curve sketching • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the nonzero inflection point of $f(x) = x^4 - 16x^3$.

Final Answer: $8$. $f''(x) = 12x^2 - 96x = 6x(2x - 16)$. Zero at $x = 0$ and $x = 8$; the nonzero inflection point is $x = 8$.

**18** [Challenging] • Target: Optimization problems • Cross-Lesson: 8.01 Area Between Curves

Find the nonzero inflection point of $f(x) = x^4 + 8x^3$.

Final Answer: $-4$. $f''(x) = 12x^2 + 48x = 6x(2x + 8)$. Zero at $x = 0$ and $x = -4$; the nonzero inflection point is $x = -4$.

**19** [Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

A farmer has 100 m of fencing to enclose a rectangular field with $x + 2y = 100$ (one side of length $x$ needs no fence, the other 2 side(s) of length $y$ each do). Find $y$ that maximizes the enclosed area $A = xy$.

Final Answer: $25$. $A(y) = (100 - 2y)y = 100y - 2y^2$. $A'(y) = 100 - 4y = 0 \Rightarrow y = \frac{100}{4} = 25$ m.

**20** [Challenging] • Target: Points of inflection and graph behavior • Cross-Lesson: 1.01 Limits Graphically and Numerically

A farmer has 72 m of fencing to enclose a rectangular field with $x + 2y = 72$ (one side of length $x$ needs no fence, the other 2 side(s) of length $y$ each do). Find $y$ that maximizes the enclosed area $A = xy$.

Final Answer: $18$. $A(y) = (72 - 2y)y = 72y - 2y^2$. $A'(y) = 72 - 4y = 0 \Rightarrow y = \frac{72}{4} = 18$ m.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Increasing/decreasing and the first derivative test

Classify the critical point $x = -1$ of $f(x) = x^3 + 6x^2 + 9x$, using the Second Derivative Test.

**a.** a local maximum

**b.** cannot be determined without more information

**c.** **a local minimum ✓**

**d.** an inflection point

$f''(x) = 6x + 12$, so $f''(-1) = -6 + 12 = 6$, which is positive: $x = -1$ is a local minimum.

**22** [Foundational] • Target: Concavity and the second derivative test

Classify the critical point $x = 1$ of $f(x) = x^3 + 3x^2 - 9x$, using the Second Derivative Test.

**a.** cannot be determined without more information

**b.** **a local minimum ✓**

**c.** an inflection point

**d.** a local maximum

$f''(x) = 6x + 6$, so $f''(1) = 6 + 6 = 12$, which is positive: $x = 1$ is a local minimum.

**23** [Foundational] • Target: Curve sketching

Classify the critical point $x = -6$ of $f(x) = x^3 + 12x^2 + 36x$, using the Second Derivative Test.

**a.** a local minimum

**b.** **a local maximum ✓**

**c.** an inflection point

**d.** cannot be determined without more information

$f''(x) = 6x + 24$, so $f''(-6) = -36 + 24 = -12$, which is negative: $x = -6$ is a local maximum.

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**24** [Foundational] • Target: Optimization problems

Classify the critical point $x = 0$ of $f(x) = x^3 - 9x^2$, using the Second Derivative Test.

**a. a local maximum ✓**

**b.** an inflection point

**c.** cannot be determined without more information

**d.** a local minimum

$f''(x) = 6x - 18$, so $f''(0) = 0 - 18 = -18$, which is negative: $x = 0$ is a local maximum.

**25** [Foundational] • Target: Optimization: minimizing cost or distance

Classify the critical point $x = 6$ of $f(x) = x^3 - 3x^2 - 72x$, using the Second Derivative Test.

**a.** cannot be determined without more information

**b.** an inflection point

**c.** a local maximum

**d. a local minimum ✓**

$f''(x) = 6x - 6$, so $f''(6) = 36 - 6 = 30$, which is positive: $x = 6$ is a local minimum.

**26** [Foundational] • Target: Points of inflection and graph behavior

Classify the critical point $x = 4$ of $f(x) = x^3 - 9x^2 + 24x$, using the Second Derivative Test.

**a.** a local maximum

**b.** an inflection point

**c. a local minimum ✓**

**d.** cannot be determined without more information

$f''(x) = 6x - 18$, so $f''(4) = 24 - 18 = 6$, which is positive: $x = 4$ is a local minimum.

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**27** [Foundational] • Target: Visualizing extrema and concavity together

Classify the critical point $x = -6$ of $f(x) = x^3 + 9x^2$, using the Second Derivative Test.

**a.** an inflection point

**b.** a local minimum

**c.** **a local maximum ✓**

**d.** cannot be determined without more information

$f''(x) = 6x + 18$, so $f''(-6) = -36 + 18 = -18$, which is negative: $x = -6$ is a local maximum.

**28** [Foundational] • Target: A capstone optimization problem

Classify the critical point $x = -5$ of $f(x) = x^3 + 3x^2 - 45x$, using the Second Derivative Test.

**a.** an inflection point

**b.** **a local maximum ✓**

**c.** cannot be determined without more information

**d.** a local minimum

$f''(x) = 6x + 6$, so $f''(-5) = -30 + 6 = -24$, which is negative: $x = -5$ is a local maximum.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Second derivative test failures

Find the nonzero inflection point of $f(x) = x^4 + 14x^3$.

**a.** $14$

**b.** $-6$

**c.** $-14$

**d.** $-7$ ✓

$f''(x) = 12x^2 + 84x = 6x(2x + 14)$. Zero at $x = 0$ and $x = -7$; the nonzero inflection point is $x = -7$.

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**30** [Intermediate] • Target: A capstone curve-sketching problem

Find the nonzero inflection point of $f(x) = x^4 - 2x^3$.

**a.** $-2$

**b.** $2$

**c.** $4$

**d.** $1$ ✓

$f''(x) = 12x^2 - 12x = 6x(2x - 2)$. Zero at $x = 0$ and $x = 1$; the nonzero inflection point is $x = 1$.

**31** [Intermediate] • Target: Concavity & Points of Inflection

Find the nonzero inflection point of $f(x) = x^4 - 14x^3$.

**a.** $8$

**b.** $14$

**c.** $7$ ✓

**d.** $-14$

$f''(x) = 12x^2 - 84x = 6x(2x - 14)$. Zero at $x = 0$ and $x = 7$; the nonzero inflection point is $x = 7$.

**32** [Intermediate] • Target: Second Derivative Test for Local Extrema

Find the nonzero inflection point of $f(x) = x^4 - 12x^3$.

**a.** $7$

**b.** $6$ ✓

**c.** $-12$

**d.** $12$

$f''(x) = 12x^2 - 72x = 6x(2x - 12)$. Zero at $x = 0$ and $x = 6$; the nonzero inflection point is $x = 6$.

**33** [Intermediate] • Target: Applied Optimization: Geometric Problems

Find the nonzero inflection point of $f(x) = x^4 - 8x^3$.

**a.** $5$

**b.** $8$

**c.** $-8$

**d.** $4$ ✓

$f''(x) = 12x^2 - 48x = 6x(2x - 8)$. Zero at $x = 0$ and $x = 4$; the nonzero inflection point is $x = 4$.

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**34** [Intermediate] • Target: Applied Optimization: Cost & Distance

A farmer has 108 m of fencing to enclose a rectangular field with $x + 2y = 108$ (one side of length $x$ needs no fence, the other 2 side(s) of length $y$ each do). Find $y$ that maximizes the enclosed area $A = xy$.

**a.** 28

**b.** 108

**c.** 27 ✓

**d.** 54

$A(y) = (108 - 2y)y = 108y - 2y^2$. $A'(y) = 108 - 4y = 0 \Rightarrow y = \frac{108}{4} = 27$ m.

**35** [Intermediate] • Target: Increasing/decreasing and the first derivative test

An open-top box is made from a 36 in by 36 in square sheet by cutting equal squares of side $x$ from each corner and folding up the sides. Find $x$ that maximizes the volume $V(x) = x(36 - 2x)^2$.

**a.** 6 ✓

**b.** 18

**c.** 9

**d.** 7

$V'(x) = (36 - 2x)^2 + x \cdot 2(36 - 2x)(-2) = (36 - 2x)[(36 - 2x) - 4x] = (36 - 2x)(36 - 6x)$. Setting $V' = 0$: $x = \frac{36}{2}$ (rejected, gives $V = 0$) or $x = \frac{36}{6} = 6$, the maximum.

**36** [Intermediate] • Target: Concavity and the second derivative test

An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side $x$ from each corner and folding up the sides. Find $x$ that maximizes the volume $V(x) = x(12 - 2x)^2$.

**a.** 4

**b.** 2 ✓

**c.** 3

**d.** 6

$V'(x) = (12 - 2x)^2 + x \cdot 2(12 - 2x)(-2) = (12 - 2x)[(12 - 2x) - 4x] = (12 - 2x)(12 - 6x)$. Setting $V' = 0$: $x = \frac{12}{2}$ (rejected, gives $V = 0$) or $x = \frac{12}{6} = 2$, the maximum.

**37** [Intermediate] • Target: Curve sketching

An open-top box is made from a 48 in by 48 in square sheet by cutting equal squares of side $x$ from each corner and folding up the sides. Find $x$ that maximizes the volume $V(x) = x(48 - 2x)^2$.

**a.** 9

**b.** 8 ✓

**c.** 24

**d.** 12

$V'(x) = (48 - 2x)^2 + x \cdot 2(48 - 2x)(-2) = (48 - 2x)[(48 - 2x) - 4x] = (48 - 2x)(48 - 6x)$. Setting $V' = 0$: $x = \frac{48}{2}$ (rejected, gives $V = 0$) or $x = \frac{48}{6} = 8$, the maximum.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Optimization problems • Cross-Lesson: 8.01 Area Between Curves

Find the $x$-coordinate of the point on the line $y = 4x + 5$ closest to the origin, by minimizing $D = x^2 + y^2$.

**a.** $-\frac{5}{17}$

**b.** $-\frac{5}{4}$

**c.** $-\frac{20}{17}$ ✓

**d.** $\frac{20}{17}$

$D(x) = x^2 + (4x + 5)^2 = (17)x^2 + 40x + 25$. $D'(x) = 34x + 40 = 0 \Rightarrow x = \frac{-20}{17} = -\frac{20}{17}$.

**39** [Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Find the $x$-coordinate of the point on the line $y = x + 3$ closest to the origin, by minimizing $D = x^2 + y^2$.

**a.** $-\frac{3}{2}$ ✓

**b.** $-3$

**c.** $\frac{3}{2}$

**d.** $-\frac{1}{2}$

$D(x) = x^2 + (x + 3)^2 = (2)x^2 + 6x + 9$. $D'(x) = 4x + 6 = 0 \Rightarrow x = \frac{-3}{2} = -\frac{3}{2}$.

**40** [Challenging] • Target: Concavity & Points of Inflection • Cross-Lesson: 1.01 Limits Graphically and Numerically

On which interval is the function $f(x) = x^3 - 6x^2 + 9x$ concave up?

**a.** $(2, \infty)$ ✓

**b.** $(-\infty, 2)$

**c.** $(1, 3)$

**d.** $(0, \infty)$

$f'(x) = 3x^2 - 12x + 9$. $f''(x) = 6x - 12$. Concave up where $f''(x) > 0 \Rightarrow 6x - 12 > 0 \Rightarrow x > 2$.

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**41** [Challenging] • Target: Concavity & Points of Inflection • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

What is a necessary condition for $(c, f(c))$ to be a point of inflection?

**a.** $f''(c) = 0$ **or** $f''(c)$ **undefined, AND** $f''$ **changes sign across** $c$ ✓

**b.** $f'(c) = 0$

**c.** $f''(c) = 0$ only

**d.** $f(c) = 0$

A point of inflection requires the concavity to change sign, meaning $f''$ must change sign across $c$ (where $f''(c) = 0$ or does not exist).

**42** [Challenging] • Target: Second Derivative Test for Local Extrema • Cross-Lesson: 2.01 The Definition of the Derivative

When is the Second Derivative Test inconclusive at a critical point $c$?

**a.** **When** $f''(c) = 0$ **or** $f''(c)$ **does not exist** ✓

**b.** When $f''(c) > 0$

**c.** When $f'(c) = 0$

**d.** When $f(c) = 0$

If $f''(c) = 0$ or fails to exist, the test provides no information and one must use the First Derivative Test instead.

**43** [Challenging] • Target: Applied Optimization: Geometric Problems • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

A farmer has $200$ meters of fencing to enclose a rectangular field adjacent to a straight river (no fence needed along the river). What dimensions maximize the area?

**a.** **Length** $100$ **m, Width** $50$ **m** ✓

**b.** Length $80$ m, Width $60$ m

**c.** Length $120$ m, Width $40$ m

**d.** Square of side $66.7$ m

Fence: $2W + L = 200 \Rightarrow L = 200 - 2W$. Area $A(W) = W(200 - 2W) = 200W - 2W^2$.
$A'(W) = 200 - 4W = 0 \Rightarrow W = 50$ m, $L = 100$ m.

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**44** [Challenging] • Target: Applied Optimization: Geometric Problems • Cross-Lesson: 3.01 The Chain Rule

What is the maximum volume in the box problem above?

**a.** 128 **cm**$^3$ ✓

**b.** 144 cm$^3$

**c.** 108 cm$^3$

**d.** 96 cm$^3$

$V(2) = 2(12-4)^2 = 2(8^2) = 2(64) = 128$ cm$^3$.

**45** [Challenging] • Target: Applied Optimization: Cost & Distance • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

A company determines that total cost for producing $x$ units is $C(x) = 0.01x^2 + 4x + 100$. What production level minimizes average cost $\bar{C}(x) = \frac{C(x)}{x}$?

**a.** $x = 100$ ✓

**b.** $x = 50$

**c.** $x = 200$

**d.** $x = 400$

$\bar{C}(x) = 0.01x + 4 + \frac{100}{x}$. $\bar{C}'(x) = 0.01 - \frac{100}{x^2} = 0 \Rightarrow x^2 = \frac{100}{0.01} = 10,000 \Rightarrow x = 100$.

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